

Helm Charts for Beginners (Kubernetes / OpenShift) — simple introduction

1) What Helm is (plain language)

Think of **Helm** as an “app installer” for Kubernetes.

- Kubernetes runs your app using YAML files (Deployment, Service, etc.).
- A **Helm chart** is a **bundle** of those YAML files + a way to customize them.

Instead of managing 15 YAML files by hand, you say:

- “Install this chart”
- “Use these settings (values)”

2) Chart structure (what’s inside)

Helm chart structure (what’s inside a chart folder)

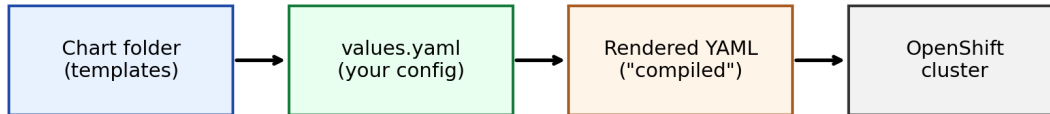
```
mychart/  
  Chart.yaml      (name/version)  
  values.yaml     (your settings)  
  templates/  
    deployment.yaml  
    service.yaml  
    route.yaml    (OpenShift)  
  charts/         (optional deps)
```

In simple terms:

- `Chart.yaml`: chart name + version.
- `values.yaml`: your settings (image, replicas, ports...).
- `templates/`: YAML templates (they contain placeholders like `{{ .Values.replicaCount }}`).

3) The Helm workflow (the big picture)

Helm workflow: chart → rendered YAML → cluster



Helm does two jobs: 1) render templates to final YAML 2) apply that YAML to the cluster + store a release record

Your colleague said “compile a Helm chart” — Helm doesn’t compile like Java.

In Helm, “compile” usually means:

- **render templates → produce final YAML**

You can render without installing:

```
``bash helm template myapp ./mychart -f my-values.yaml ``
```

4) Common commands (cheat sheet)

```
``bash
```

Create a starter chart

```
helm create mychart
```

Validate the chart

```
helm lint ./mychart
```

Render ("compile") to YAML without a cluster

```
helm template myapp ./mychart -f my-values.yaml
```

Install

```
helm install myapp ./mychart -n my-namespace -f my-values.yaml
```

Upgrade

```
helm upgrade myapp ./mychart -n my-namespace -f my-values.yaml
```

Rollback

```
helm rollback myapp 1 -n my-namespace
```

Uninstall

```
helm uninstall myapp -n my-namespace ``
```

5) “How do I get a Helm chart inside the OpenShift cluster?” (simple answer)

Very important clarification:

- You usually **don’t upload the chart into the cluster** as a permanent “chart object.”
- What happens is:

1) Helm reads the chart **from your laptop / CI runner** 2) Helm renders templates to YAML 3) Helm sends YAML to the OpenShift API 4) The cluster stores a **release record** so Helm can manage it later

So the real question is:

```
“How do I run helm install against my OpenShift cluster?”
```

A) Connect Helm to OpenShift

```
``bash oc login https://api.:6443 oc new-project my-project
```

or

```
oc project my-project
```

```
helm install myapp ./mychart -n my-project -f my-values.yaml ``
```

B) OpenShift specifics (what might break)

- In Kubernetes you often use **Ingress**.
- In OpenShift you often use **Route**.
- Some charts need small changes to work on OpenShift (security/permissions too).

6) How teams store/share charts (3 common ways)

How teams store/share Helm charts (3 common ways)



CI/CD typically runs: `helm upgrade --install ...` (from Git, repo, or OCI)

- **Git repo**: simplest (CI installs from a folder)
- **Helm repo**: package `.tgz` + `index.yaml` and host it
- **OCI registry**: store charts in a registry (like images)

7) Your case: JFrog (Artifactory) + OpenShift (simple call flow)
 If your charts live in JFrog Artifactory, the key point is:
 - Your laptop/CI pulls the chart from JFrog
 - Helm renders YAML locally
 - Helm applies YAML to the OpenShift API
 - Cluster nodes pull container images, not the Helm chart
 Diagram: (helm-assets/jfrog_helm_openshift_callflow.png)